

Chunping (Fred) Gao

BEng, PhD, MASHRAE, MIES, GSAP

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Summary

Experienced engineering and building science specialist with 20 years of expertise delivering healthy and sustainable infrastructure projects across public sector and consulting environments in Australia and Asia. Specialised in the design, operation, and risk management of building services systems, with a particular focus on healthcare infrastructure. Proven ability to manage air and water system risks to support safe, healthy, and high-performing built environments. Skilled in providing evidence-based technical advice through research, modelling, field measurements, and comprehensive reporting. Extensive experience in incident investigation, policy and guideline development, and stakeholder engagement. Recognised for strong analytical and problem-solving capabilities, including the assessment and implementation of emerging technologies, with multiple industry awards and recognitions.

Key Skills

Technical & Regulatory Expertise

- **Building Services Consultancy:** Design and operational consultancy for building mechanical, electrical and plumbing (MEP) systems.
- **System Analysis & Risk Assessment:** Infrastructure performance analysis and risk assessment using advanced tools and methodologies, including CFD, energy modelling, and GIS.
- **Site Verification & Post-Occupancy Evaluation:** Field measurement and verification (M&V), environmental sampling, auditing, and post-occupancy performance evaluation.
- **Codes, Standards & Guidelines:** Strong working knowledge of the National Construction Code (NCC), Australasian Health Facility Guidelines (AusHFG), AIRAH, ASHRAE, CIBSE, WELL, and Green Star frameworks, as well as government processes and policy requirements.

Analytical, Problem-Solving & Stakeholder Engagement Skills

- Troubleshooting and resolution of building services issues, including system underperformance, indoor air and water quality, mould growth, and infection control risks.
 - Research and analysis of standards, guidelines, and best practices for healthcare facility design and operations, including airflow, moisture and heat transfer, and acoustics.
 - Evaluation of technical challenges and emerging technologies related to building services systems and environmental control.
 - Preparation of clear and concise technical reports, advice, and recommendations for government, consultants, and contractors.
 - Facilitation of cross-disciplinary workshops, design reviews, and stakeholder engagement activities to support project delivery and decision-making.
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Professional Experience

Senior Scientific Officer

Department of Health, Western Australia (Apr 2026 - Present)

- Provides expert advice and makes recommendations in relation to air handling and water distribution system management in buildings.
- Undertake complex operational activities and technical environmental health assessments related to air handling and water distribution systems in buildings.
- Support Health Service Providers to apply and comply with a range of requirements contained with Australian Standard AS3666 and the Australian Drinking Water Guidelines.

Principal Building Research Officer

Department of Housing and Works, Western Australia (Nov 2023 – Nov 2025)

- Led WA's climate adaptation building research initiative, overseeing project management and core main research activities, including case study development, risk management tools creation and guidelines development.
- Delivered advanced building modelling and HVAC performance analysis to inform building services' design solutions responsive to climate and environmental change.
- Provided technical evaluations and design reviews for public health facilities and schools across WA, ensuring compliance, performance and risk-mitigation outcomes.

Key Achievements

- Authored reports, technical guidelines and practical tools including worksheets and GIS mapping tools.
- Developed prototype models for typical building typologies, including public schools, hospitals, and general office buildings, to support HVAC design review and optimisation.
- Delivered design reviews for critical government projects, including Tom Price and Paraburdoo Hospitals.

Senior Consultant

ARUP, Perth (Oct 2022 – Nov 2023)

- Delivered environmental design and consultancy services, including the preparation of drawings, specifications, site inspection and performance verification, to achieve healthy building certification under WELL standard.
- Led modelling and design optimisation for complex HVAC systems, incorporating strategies to improve performance of air handling units (AHU) and cooling towers (CT).
- Liaised with clients and stakeholders to ensure project delivery, conducted design workshops and reviews, provided HVAC coordination throughout construction and post occupancy evaluation phases.

Key Achievements

- Delivered WELL certifications for multiple commercial projects including **Elizabeth Quay Lot 6, Westralia Square 2** (Engineers Australia Outstanding Project Award 2024).
- Delivered HVAC modelling and design optimisation for major projects including, **St Martins Centre, Metro Martin Place, EQX SY10 Data Centre** and **Macquarie University PAEB**.
- Delivered policy report on agency clients, including Main Roads WA Net Zero Transition Plan.

Director, Built Environment Research and Innovation Institute

Building and Construction Authority (BCA) of Singapore (Mar 2015 - Oct 2022)

- Led research and deployment of advanced green building technologies under the Green Buildings Innovation Cluster (GBIC) programme, working closely with academic partners and industry stakeholders.
- Drove national initiatives on indoor air quality and health-centred building standards, contributing to the development and implementation of the health pillar within Singapore's Green Building Masterplan.
- Managed high-performance buildings, including the BCA Zero Energy Building, ensuring sustained operational excellence through integrated facility management and innovative performance-optimisation strategies.
- Oversaw the delivery of demonstration green building projects to support Singapore's Green Building Masterplan.

Key Achievements

- Delivered **Green Mark 2021** including the Health & Wellbeing Section.
- Delivered HVAC inspections and incident investigations, issued risk management guidelines for reducing **COVID-19** virus transmission risks in buildings.
- Delivered over 20 demonstration sustainable building projects, including Keppel Bay Tower, United World College, DBS Newton Office, SMU Connexion, etc.
- Contributed to World Green Building Council's framework on health and wellbeing and **TR114:2023 Mechanical electrical and plumbing (MEP) design for healthcare facilities**.

Senior Development Officer, Green Mark Department

Building and Construction Authority of Singapore (Mar 2008 - Oct 2015)

- Led development and implementation of Singapore's Green Building rating system Green Mark, including design assessment and post-occupancy verification.
- Provided design review and expert advice to public sector agencies on green solutions and HVAC optimisation in Singapore's tropical context.
- Undertook research and analysis to support policy formulation and execution.

Key Achievements

- Assessed more than 200 building projects in achieving Green Mark certification, with a focus on HVAC performance.
- Reviewed and provided expert advice on significant public works projects, including Singapore Sports Hub, Gardens by the Bay, Changi Airport Terminal 3 & 4, Jewel, Marina Barrage, etc.
- Contributed to the delivery of building codes, SS530, SS553, SS554 and Green Mark (equivalent to NCC Section F&J, AS1668, NABERS and Green Star)

Senior Consultant

BCA International Ptd Ltd (secondment from BCA, Mar 2008 - Mar 2015)

- Delivered mechanical engineering and green building consulting services in compliance with international green building standards (WELL, LEED, Green Mark), specially on HVAC airside systems and hydronic plant design optimisation.
- Conducted detailed site inspection and performance verification, including IAQ and energy audits.

Key Achievements

- Developed green building design standards and technical guidelines for government-to-government collaborations in Tianjin, Suzhou, and Guangzhou.
- Delivered multiple high-profile green building projects in Tianjin Eco-city, including the National Animation Park, Low Carbon Living Lab, and Eco-business Park.
- Delivered UNEP project on Nationally Appropriate Mitigation Action (NAMA) Development for the building sector in Asia, supporting climate mitigation planning across the region.

Research Fellow

National Neuroscience Institute, Singapore (Aug 2006 - Mar 2008)

- Conducted research applying biomedical engineering principles to support clinical innovation and improve patient-care technologies.
- Supported upgrades to health infrastructure to enable the deployment and integration of advanced medical equipment and clinical systems.

Adjunct Associate Professor

Nanyang Technological University, Singapore (May 2017 - Mar 2019)

- Led research and analysis on building energy policies and renewable energy deployment, providing evidence-based insights to inform strategic planning and regulatory development.
- Provided technical guidance and mentorship to junior research engineers, supporting their professional development and the quality of project outcomes.

Achievements:

- ▶ Delivered testing of green technologies at BCA SkyLab in collaboration with Berkeley Lab.
- ▶ Delivered IEA-EBC Annex 81: Data-driven smart buildings, as a subtask leader.

Building Services Engineer

Zhejiang Institute of Building Research, China (Jul 1997 - May 2002)

- Prepared project concept, preliminary, and detailed designs for building services systems, with a focus on advanced mechanical and hydraulic solutions.
- Performed commissioning, verification and performance measurements of building services systems to ensure compliance and delivery of intended performance outcomes.

Key Achievements

- Delivered HVAC solutions for Xiaoshan International Airport Terminal 1, Grand Hyatt Westlake Hotel, Yongjin Mall and Zhejin Plaza.

Education

- **PhD (Biomedical Engineering)** - National University of Singapore (2007)
 - **Research Scholar** in building physics, Carnegie Mellon University (2015)
 - **Postgraduate Diploma** in building physics, Zhejiang University (2002)
 - **B.Eng** in Mechanical and Electrical Engineering, Harbin University of Civil Engineering and Architecture (1997)
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Qualifications

- Member of American Society of Heating, Refrigeration and Air-conditioning Engineers (ASHRAE)

- Member of Institution of Engineers Singapore (MIES)
 - Green Star Accredited Professional (GSAP)
 - LEED Accredited Professional (LEED AP)
 - Green Mark Accredited Professional (GMAP)
 - Certified Project Manager Level I, Arup
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Professional Training

- Green Star Foundation and Advanced Courses, GBCA (2014, 2024)
 - Building Simulation/Analysis Using IESVE, IESVE (2015, 2022, 2025)
 - HVAC Training Courses on Services and Controls, Carrier (2010, 2015)
 - Performance Measurement and Verification with IPMVP, EVO (2018)
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Awards & Recognition

- National Awards - COVID-19 Resilience Medal (Team & Individual) - Public Service Commission of Singapore, Sep 2023
 - Prestigious Engineering Achievement Award, Institution of Engineers, Singapore (Aug 2017)
 - SG50 Top 50 Engineering Feats Award, Institution of Engineers, Singapore (Jul 2016)
 - Prestigious Engineering Achievement Award for SkyLab Institution of Engineers, Singapore (Aug 2017)
 - SG50 Top 50 Engineering Feats Award for ZEB, Institution of Engineers, Singapore (Jul 2016)
 - MND Minister's Award, Ministry of National Development, Singapore (Aug 2017)
 - BCA Exemplary Team Award (2017-2022), Building and Construction Authority, Singapore
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Selected Publications

- Possible inter-room aerosol-mediated COVID-19 transmission in a quarantine hotel: A case study, **Building and Environment**, Volume 295, 2026
 - **IEA-EBC Annex 81 'Data-Driven Smart Buildings' Technical Report**, 2025
 - Energy Savings and Thermal Comfort in a Zero Energy Office Building with Fans in Singapore, **Building and Environment**, Volume 243, 2023
 - IEQ Applications: Living Lab integrates IEQ Technologies in Singapore, **ASHRAE Journal**, Feb 2020
 - Health, Wellbeing and Productivity in Offices: the next chapter for green building, WorldGBC, Sept 2014
 - Investigation of an Integrated Automated Blinds and Dimmable Lighting System for Tropical Climate in a Rotatable Testbed Facility, *Energy and Buildings*, Vol 183, Jan 2019
 - Evaluation of Thermal Comfort Criteria of an Active Chilled Beam System in Tropical Climate, *Building and Environment*, Vol 145, Nov 2018
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References

Available upon request